

## Power Consumption Analysis:

Power analysis is used to understand the energy requirement of the STM32, sensors and ESP32 communication module. Measurements should be taken during different operating conditions.

| State              | Power Strategy                   | Measurement to Record   |
|--------------------|----------------------------------|-------------------------|
| STM32 Monitoring   | Efficient periodic sensing       | Voltage, current, power |
| Sensor Measurement | Avoid unnecessary sampling       | Sensor current          |
| ESP32 Wi-Fi        | Optimize communication           | Wi-Fi current           |
| Normal Operation   | Monitor complete system          | Total current/power     |
| Alert Condition    | Observe additional communication | Peak current            |
| Wi-Fi Recovery     | Check reconnection load          | Peak current            |

## Reliability Testing:

Reliability testing verifies whether the system continues to operate correctly during repeated operation, communication interruptions and repeated alert conditions.

| Test Area     | Reliability Check             | Acceptance Focus    |
|---------------|-------------------------------|---------------------|
| IV Monitoring | Repeated weight measurements  | Stable readings     |
| UART          | Continuous communication      | No corrupted frames |
| Wi-Fi         | Repeated disconnect/reconnect | Successful recovery |

|                |                              |                          |
|----------------|------------------------------|--------------------------|
| Alerts         | Repeated threshold crossing  | Correct alert every time |
| Priority Queue | Multiple simultaneous alerts | Consistent priority      |
| Nurse Controls | Repeated ACK/Solve           | Correct state transition |
| Long Operation | Continuous system operation  | No unexpected failure    |

## Future Improvements

The following improvements can increase the practical reliability and scalability of the system:

- **Persistent alert history** using non-volatile/cloud storage.
- **Multiple physical beds** instead of a single prototype bed.
- **Communication timeout detection** for stale STM32 data.
- **Battery backup** for short-duration power interruptions.
- **Improved FSR arrangement** to reduce false bed-exit detection.
- **Mobile notifications** for critical alerts.
- **Encrypted communication** for secure patient-related data.
- **Cloud database integration** for long-term monitoring and analysis.

## Overall Test Summary

Phase 5 validates the complete system from **sensor acquisition to nurse alert management**. Testing covers functional operation, performance, communication reliability, fault handling, power requirements and system recovery.

## Conclusion

The Testing and Validation Review confirms that the proposed Smart Hospital Care System can integrate **IV monitoring, bed-exit detection, real-time communication and nurse-oriented alert management** into a single prototype.

The testing process also identifies areas requiring further improvement, particularly **communication recovery, power optimization, multi-bed scalability and long-term data storage**. The results obtained from Phase 5 will provide the basis for improving the system's reliability and preparing it for future multi-bed implementation.